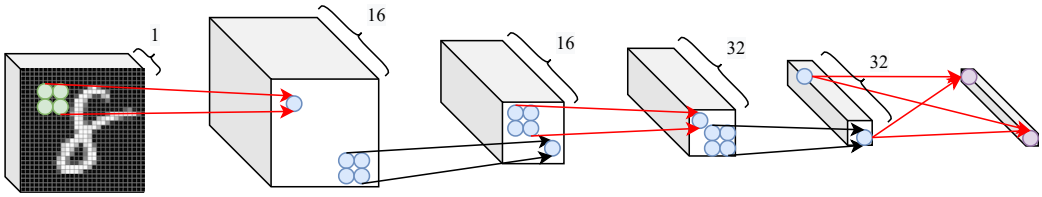
Neural Systems

(Academic year 2024-2025)

Exercises on supervised learning: Training a network with two hidden convolutional layers

Exercise 12

Handwritten digit classification is a common benchmark task addressed when designing deep neural networks with gray-scale images as input. The task consists in associating to a handwritten digit the correct class, i.e., $L = \{"0", \dots, "9"\}$ (10-way classification task). The most used dataset is MNIST (http://yann.lecun.com/exdb/mnist/), composed by 60000 training and 10000 test images 28x28.



Input layer: layer consisting in neurons organised in a 28x28x1 grid (replicating the input image), standardizing the input using training statistics (mean and standard deviation).	**Hidden layer 0 (+activation)**: convolutional layer learning 16 filters of size (5,5), activated via ReLU.	**Pooling layer 0 (+dropout)**: average pooling layer with pooling size of (2,2) and stride of (2,2), followed by dropout (dropout rate at 0.25).	**Hidden layer 1 (+activation)**: convolutional layer learning 32 filters of size (5,5), activated via ReLU.	**Pooling layer 1 (+dropout)**: average pooling layer with pooling size of (2,2) and stride of (2,2), followed by dropout (dropout rate at 0.25).	**Output layer (+activation)**: fully-connected layer with 10 neurons (10 way-classification), activated via Softmax.
Number of neurons: Number of trainable parameters (weights+bias): Output size:	Number of neurons: Number of trainable parameters (weights+bias): Output size:	Number of neurons: Number of trainable parameters (weights+bias): Output size:	Number of neurons: Number of trainable parameters (weights+bias): Output size:	Number of neurons: Number of trainable parameters (weights+bias): Output size:	Number of neurons: Number of trainable parameters (weights+bias): Output size:

- Data pre-processing and split. As point 1 of exercise 11
- Network design. As point 2 of exercise 11. Use the proper functionalities to define convolutional and pooling layers.
- Network training. As point 3 of exercise 11.
- Network evaluation. As point 4 of exercise 11. Compare the performance on the test set between the fully-connected network (exercise 11) and the convolutional network.
- Network evaluation on your own digits. Write one digit (see 'sample_digit.jpg') on a white paper and take a picture; transfer it on your computer. Use the function provided in 'load_my_digit.py' to load your image and process it to be compliant with MNIST images. Test the network on your digit and get the predicted top-3 classes.
- Feature visualization. Extract the convolutional kernels learned at the first convolutional layer and visualize them. What kind of features were exploited? To further support your interpretation, you can also visualize the output of the first convolutional layer, by computing the 2-D convolution between a representative input (e.g., your digit loaded at the previous point) and each learned kernel.

Notes. The needed Python functionalities needed to solve the exercise are described in the lecture note "Keras for supervised learning: An introduction". In case the training loop takes some time on your computer, you can use the free online service Google Colaboratory (enabling GPU in the "Edit→Notebook Settings" tab) to train your models (see https://colab.research.google.com/).